

SIMATS ENGINEERING
Department of Computer Science & Engineering
ASSESSMENT TOOL 1- EXPERIMENTS

Course Code:	CSA1223	Course Title:	Computer Architecture
CO Assessed:	CO3	Assessment:	ASSESSMENT TOOL 1- EXPERIMENTS
Weightage:	40%	Total Marks:	100
CO3 Statement:	<i>Design processor organization by constructing pipelined datapath structures, identifying hazard types (data, control, structural), and comparing superscalar techniques such as out-of-order execution, speculative execution, and simultaneous multithreading (SMT). (BL6)</i>		
Student Name:	VARSHA VARDHINI. S	Reg.No.:	192511488
Department:	BE - CSE	Date:	

ANNEXURE A

1. Design and simulate a system that performs register transfer operations along with basic arithmetic and logic operations such as addition, subtraction, AND, and OR. Use multiple registers to demonstrate how data is transferred between them during execution, and analyze the sequence of operations and the role of the ALU in processing data.
2. Create a model of a computer system using single-bus and multi-bus organization, and perform data transfer operations between registers, memory, and I/O. Compare the time taken and efficiency of both approaches, and analyze how multiple bus structures improve parallel data transfer and reduce bottlenecks.
3. Simulate a 5-stage instruction pipeline (Fetch, Decode, Execute, Memory, Write-back) and execute a set of instructions. Measure the performance in terms of speedup and throughput, and compare it with a non-pipelined system to evaluate the effectiveness of pipelining.
4. Develop a pipeline execution scenario where data hazards, control hazards, and structural hazard occur. Observe their impact on instruction execution, and implement techniques such as stalling, data forwarding, and branch prediction to resolve these hazards and improve performance.
5. Analyze a processor supporting superscalar execution, including out-of-order and speculative execution, and extend the study to include hyper-threading and simultaneous multithreading (SMT). Evaluate how multiple instructions are executed in parallel and compare the performance improvements in terms of instruction throughput and resource utilization.

Code of Conduct:

I VARSHA VARDHINI-S /192511488 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: S. Varsha

MARKS ALLOCATION

	Understanding of Concepts (20%)	Experimental Design (20%)	Use of Tools (25%)	Analysis of Results (20%)	Documentation (15%)	
Q1	4	5	3	3	2	17
Q2	4	4	2	2	1	13
Q3	3	4	2	2	1	12
Q4	3	3	3	3	1.5	13.5
Q5	4	2	2	2	2	12

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies(5)	Good understanding with proper integration of most concepts(4)	Basic understanding; limited or partial integration(3)	Poor understanding; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution(6)	Correct implementation with minor issues(4)	Basic implementation with errors or inefficiencies(3)	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration(4)	Appropriate use of tools with correct execution(3)	Limited or inconsistent use of tools(2)	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results(3)	Good analysis with reasonable interpretation(2)	Basic analysis with limited insights(1)	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results(2)	Organized documentation with necessary details(1.5)	Basic documentation with missing details(1)	Poor or incomplete documentation

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ANNEXURE A

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ANSWER :

Aim

To design and simulate a system using multiple registers to perform register transfer operations and basic arithmetic and logic operations such as **addition, subtraction, AND, and OR** using the **8085 microprocessor simulator (GNUSim8085)**.

Understanding of Concept

In an 8085 microprocessor, registers are temporary storage locations used to hold data during program execution.

For this experiment:

- **B Register** → First operand
- **C Register** → Second operand
- **A Register** → Used by the ALU to perform operations
- **D Register** → Stores/transfers the addition result
- **E Register** → Stores/transfers the subtraction result
- **Memory locations** → Store the results for verification

The ALU performs:

- Addition
- Subtraction
- AND
- OR

Experimental Design

Operation	Expression	Expected Result
Addition	0AH + 06H	10H
Subtraction	0AH – 06H	04H
AND	0AH AND 06H	02H
OR	0AH OR 06H	0EH

Tool Used

GNUSim8085 – 8085 Microprocessor Simulator

The tool is used to:

1. Enter the 8085 assembly program.
 2. Assemble the program.
 3. Execute the instructions.
 4. Observe register values.
 5. Observe memory locations.
 6. Capture the output screenshot.
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2. Create a model of a computer system using single-bus and multi-bus organization, and perform data transfer operations between registers, memory, and I/O. Compare the time taken and efficiency of both approaches, and analyze how multiple bus structures improve parallel data transfer and reduce bottlenecks.
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